

CSSE 220 – Object-Oriented Software Development
Rose-Hulman Institute of Technology

Worksheet 11

Name (Print): _____ Section: _____

1. Understanding this in Java:

- `this.field` → access current object's data
- `this.method()` → call current object's method
- `this(...)` → call another constructor (same class)

For each line below, choose what this is doing:

- A.** Access the current object's data
- B.** Call the current object's method
- C.** Call another constructor from the same class

```
1 class Drink {  
2     String name;  int calories;  
3     public Drink(String name, int calories) {  
4         this.name = name;  
5         this.calories = calories;  
6     }  
7     public Drink(String name) {  
8         this(name, 0);  
9     }  
10    public void printInfo() {  
11        this.display();  
12    }  
13    public void display() {  
14        System.out.println(this.name + " " + this.calories);  
15    }  
}
```

- (a) `this(name, 0);` → _____
- (b) `this.name = name;` → _____
- (c) `this.display();` → _____
- (d) `this.name` → _____

2. What happens if we move this(...) to the second line?

```
1 public Drink(String name, int calories) {  
2     System.out.println("Creating drink...");  
3     this(name, calories, false);  
4 }
```

- (a) OK

- (b) Runtime error
- (c) Compile-time error

3. **Relationships.** Select from the following: **IS-A, HAS-A, DEPENDS-ON**

- (a) class Car implements Drivable: _____
- (b) class Car { private Engine engine; }: _____
- (c) class Vet { void examine(Dog d) {...}}: _____

4. **Definition (Inheritance)**

Inheritance is a mechanism when a _____ automatically gains the fields and methods of its _____ by using the **extends** keyword.

5. **Java class hierarchy**

What is the name of the topmost class in Java? _____

6. **What can a subclass do? (circle/select all correct)**

- (a) A subclass inherits fields and methods from its superclass, but private members are not directly accessible.
- (b) A subclass inherits constructors from its superclass.
- (c) A subclass must call (explicitly or implicitly) a superclass constructor.
- (d) A subclass can override public and protected methods (if they are not **final** or **static**).
- (e) A subclass can override private methods.

7. **Using super.** Fill in the correct Java code.

- (a) Call a superclass method **eat()**: _____
- (b) Call a superclass constructor **Animal(String name)**: _____

8. **Overriding: what prints?.** Write the output for each line:

```

1 class Animal {
2     void move() { System.out.println("Animal is moving."); }
3     void eat() { System.out.println("Animal is eating."); }
4 }
5 class Dog extends Animal {
6     @Override
7     void move() {System.out.println("Dog is running."); }
8     void bark() { System.out.println("Dog is barking."); }
9 }
10 public class Main {
11     public static void main(String[] args) {
12         Dog d = new Dog();
13         d.move();    // Output 1: -----
14         d.eat();     // Output 2: -----
15         d.bark();    // Output 3: -----
16     }}
```

9. **Before You Leave**

Write one question you still have about today's topic.